Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	5 March 2026
Team ID	Team 25
Project Name	Quotes Recommendation Chatbot using NLP
Maximum Marks	4 Marks

Technical Architecture:

S.No	Component	Description	Technology
1	User Interface	Interface where users interact with the chatbot to request quotes or enter moods/keyw	HTML, CSS, JavaScript / React JS
2	Application Logic-1	Handles user registration, login authentication, and request processing.	Python / Flask
3	Application Logic-2	Processes user input using NLP techniques such as tokenization and keyword extraction.	Python NLP Libraries (NLTK / spaCy)
4	Application Logic-3	Recommendation logic that matches user input with relevant quotes.	Python, Scikit-learn
5	Database	Stores quotes, authors, categories, and user data.	MySQL / SQLite
6	Cloud Database	Cloud-based storage for scalable data management.	MongoDB Atlas / IBM Cloudant
7	File Storage	Storage of datasets, logs, and application files.	Local File System / Cloud Storage
8	External API-1	API used to fetch additional quotes or related content.	Quotes API / ZenQuotes API
9	External API-2	API used for authentication or external services.	Google OAuth API

10	Machine Learning Model	NLP model used to understand user input and recommend relevant quotes.	TF-IDF / NLP Recommendation Mode
11	Infrastructure (Server / Cloud)	Deployment environment for hosting the chatbot application.	Local Server / AWS / IBM Cloud

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	List of frameworks used to build the chatbot and NLP system.	Python, Flask, NLTK, spaCy
2	Security Implementations	Security measures for user authentication and data protection.	Password Encryption, OAuth, HTTPS
3	Scalable Architecture	Architecture that allows the system to support increasing users and data.	3-Tier Architecture, Cloud Deployment
4	Availability	Ensures the application is accessible with minimal downtime.	Cloud Hosting, Load Balancing
5	Performance	Optimizes system response time and efficiency.	Caching, Optimized Database Queries